

Spectral Analysis of The Pleiades Cluster

Hanaan Khan

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Abstract

During the course of Astrolab, we analysed CCD images of the Pleiades cluster using data from our very own Michael Gardner Observatory. We measured the telescope’s plate scale by comparing observed pixel separations of Atlas and Pleione to their known angular separation. Using image analysis and Gaussian fitting, we estimated the point spread function (PSF) to ascertain the telescope’s angular resolution under the observing conditions present at the time. Finally, aperture photometry was utilised to measure the relative fluxes and magnitudes of several stars in the field, allowing us to compare our data to already established stellar magnitudes. Our findings demonstrate a consistent plate scale of 0.61 ± 0.01 arcsec/pixel, a PSF width of roughly 3.4 ± 0.1 pixels, and reliable magnitude differences across the chosen stars. These results show us the effectiveness of ground-based observations from the university when coupled with rigorous data analysis, and highlight the sensitivity of CCD measurements to both the instrumental and atmospheric effects. The methods shown are directly applicable to the wider aspects of astronomical studies, such as stellar classification or the making of Hertzsprung-Russell diagrams.

1 Introduction

Astronomy in general relies on the precise measurements of light from celestial objects to understand physical properties such as position, temperature, brightness and distance. In Astrolab, we used CCD imaging of the Pleiades open star cluster to explore three key aspects of astronomical data analysis: plate scale, point spread function (PSF), and aperture photometry.

The data was obtained using the Michael Gardner Observatory at Royal Holloway, University of London. The observatory contains a 30 cm Meade LX200 Schmidt-Cassegrain telescope with a focal length of $F = 3048$ mm, with a SBIG ST-8XME CCD camera attached. The camera’s chip consists of 1530×1020 pixels, each $9 \times 9 \mu\text{m}$ in size, producing high-resolution images suitable for our analysis.

Our analysis is based on the Pleiades cluster, an open cluster located approximately 135 parsecs from Earth. Its stars are relatively bright and closely spaced making it an ideal target for photometric and spatial analysis. The two stars we focused on were stars such Atlas and Pleione, which are well-separated yet close enough to fit within a single CCD frame.

This project was structured into three main analyses. First, we determined the plate scale in arcseconds per pixel by comparing the pixel separation between known stars to their angular separation based on SIMBAD data. Next, we investigate the PSF by modelling the brightness profile of stars using Gaussian fits, which helps to figure out the optical resolution of our system. Finally, we use aperture photometry to measure stellar fluxes and calculate magnitude differences, comparing our results to published

readily available data.

2 Telescope and CCD Camera

The observations for this project were carried out using the Schmidt-Cassegrain telescope located in the Michael Gardner Observatory on the rooftop of the Tolansky building. The telescope has a primary mirror diameter of 30.48 cm and a focal length of $F = 3048$ mm. This relatively long focal length, combined with the high pixel density of the CCD, allows for detailed resolution of compact star fields like the Pleiades.

Attached to the telescope is the SBIG ST-8XME CCD camera. This camera uses a CCD chip measuring 13.8×9.18 mm, with a total of 1530×1020 pixels, each $9 \times 9 \mu\text{m}$ in size. The chip operates with a typical gain factor of 2.39 photo-electrons per ADU (Analog-to-Digital Unit), and has a peak quantum efficiency near 660 nm — which is perfect for observations in the visual band (V filter) [1].

The CCD exposures were saved in FITS format, allowing for direct data analysis via Python using the `astropy.io.fits` module [2]. Each FITS file includes a header containing image metadata, and a 2D array of pixel intensities representing the image. This data was analysed using the techniques outlined in the PH2260 Astrolab Week 2 materials [4].

For this experiment, a number of exposures of the Pleiades were taken through a V filter. The exposure times ranged between 2 and 20 seconds, depending on the image brightness. Files such as `atlas_pleione_2s_000.fit` and `m45_v_20s.fit` were selected for their balance of exposure and star field image crispness.

All data analysis was carried out on our own engines using Python. The main libraries used included `astropy`, `scipy.optimize`, and `matplotlib`. The DS9 viewer [3] was used to inspect images and identify star positions manually.

3 Plate Scale

We are looking at the relation between the angular separation θ between two stars and the linear separation s between their images in the focal plane of the telescope.

The quantity $d\theta/ds$ is called the *Plate Scale* P . It is related to the focal length of the telescope F by:

$$P = \frac{d\theta}{ds} = \frac{1}{F} \quad (1)$$

and Equation (1) gives us the Plate Scale in radians per unit length. If we are recording the image with a CCD camera with pixels of a certain size d , it is better to express P in units of angle (such as arcseconds) per pixel, given by:

$$P = \frac{1 \text{ rad}}{F [\text{mm}]} \times \frac{180^\circ}{\pi \text{ rad}} \times \frac{3600''}{1^\circ} = \frac{206,265''}{F [\text{mm}]} \times \frac{d [\text{mm}]}{1 \text{ pixel}} \quad (2)$$

Angular Separation between two stars separated by s pixels:

$$\theta [\text{arc} - \text{sec}] = P [\text{arc-sec/pixel}] \times s [\text{pixels}] \quad (3)$$

Focal Length: 3048 mm

Pixel Size of CCD: 9.0 μm

Therefore we can find that the Plate Scale P :

$$P = \frac{206,265}{3048} \times \frac{0.009}{1} = 0.609 \text{ arc-sec/pixel} \quad (4)$$

Image Number	Atlas (x, y)	Pleione (x, y)
Image 1	(415, 827)	(330, 384)
Image 2	(435, 839)	(385, 329)
Image 3	(440, 842)	(405, 342)
Image 4	(438, 839)	(410, 344)
Image 5	(445, 839)	(415, 344)
Image 6	(440, 842)	(410, 347)
Image 7	(440, 844)	(410, 347)
Image 8	(440, 842)	(408, 344)
Image 9	(442, 842)	(410, 347)
Image 10	(443, 844)	(413, 347)

Table 1: The measured x and y values of Atlas and Pleione

For the above table, ten images were used and the positions of both Atlas and Pleione were recorded by placing the pointer on the brightest spot. Let it be known that 'Image 1' is far removed from the values of all the other images and is most likely an outlier due to a plethora of reasons such as a shift in the telescope. Nevertheless, it will still be included in our averages.

Now that we have found the positions of both Atlas and Pleione on all ten images, we will calculate their separation and angular separation as well as their uncertainty. The uncertainty (standard deviation) is given as:

$$\sigma_x = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (5)$$

The uncertainty is given as:

$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{N}} \quad (6)$$

With Equation (5) and Equation (6) we can now calculate the uncertainty of all our measurements.

Now in order to find the Standard Deviation σ_x and Uncertainty $\sigma_{\bar{x}}$ of our Angular Separation θ , we will use the equation:

$$\sigma_{\theta} = \sigma_x \times P \quad (7)$$

in order to find the Standard Deviation. To find the Uncertainty, we utilise:

$$\sigma_{\bar{\theta}} = \sigma_{\bar{x}} \times P \quad (8)$$

In addition to any calculated uncertainty, we must also take into account systematic uncertainties. One of the sources could arise from an incorrect plate scale, where if our focal length or pixel size was measured incorrectly then in turn all our angular separations would be slightly off. Something simple such as consistently misidentifying the stars by confusing Pleione for Atlas and vice versa; incorrect scaling in the FITS files could also have been a

Image Number	Separation (pixels)	θ (arcseconds)
Image 1	451.08	274.41
Image 2	511.45	311.48
Image 3	500.86	305.02
Image 4	495.14	301.13
Image 5	496.69	302.08
Image 6	498.19	303.00
Image 7	499.80	304.01
Image 8	500.33	304.33
Image 9	500.46	304.41
Image 10	501.29	304.92
Average (Mean)	[495.53]	[301.48]
Standard Deviation	± 16.21	± 9.86
Uncertainty	± 5.13	± 3.12

Table 2: Separation between Atlas and Pleione in Pixels and Arcseconds

source of deviation for our separation. Following on from this, we can also consider optical distortions and rotation or alignment issues, all affecting the way we view the image. These problems can be used to explain the outlier in Image 1.

Comparing our findings to the accepted value will help us ascertain just how accurate our findings were. Hence, the angular separation (γ) for Atlas and Pleione was calculated utilising official data found on SIMBAD, using the published right ascension (α) and declination (δ):

$$\cos \gamma = \sin \delta_1 \sin \delta_2 + \cos \delta_1 \cos \delta_2 \cos(\alpha_1 - \alpha_2) \quad (9)$$

The result we find is consistent with the expected separation between Atlas and Pleione. The consistency across multiple images and the low uncertainty confirm the reliability of our plate scale methodology.

4 Point Spread Function

Stars appear as blurred spots in CCD images due to diffraction, atmospheric turbulence and optical imperfections. This blurring is explained by the Point Spread Function (PSF), which describes how the light from a single star is spatially distributed across the detector. A precise value of the PSF is critical for photometry, astrometry and understanding the resolving power of the system as well as general cases of spectral analysis (such as our own experiments).

For this experiment, the PSF was modelled using a one-dimensional Gaussian profile fit to the intensity values along a row of pixels passing through a selected star. This row was extracted from the image `m45_v_20s.fit` using the `astropy.io.fits` module to read the FITS file into a NumPy array. The area we were interested in was cropped manually based on the coordinates of a bright, isolated star.

The fit function used was:

$$f(x) = s \cdot \exp\left(-\frac{(x - \mu_x)^2}{2\sigma_x^2}\right) + b \quad (10)$$

where s is the signal amplitude (ADU), b is the background offset, μ_x is the centre of the star in pixel coordinates, and σ_x is the standard deviation of the profile, representing the width of the PSF.

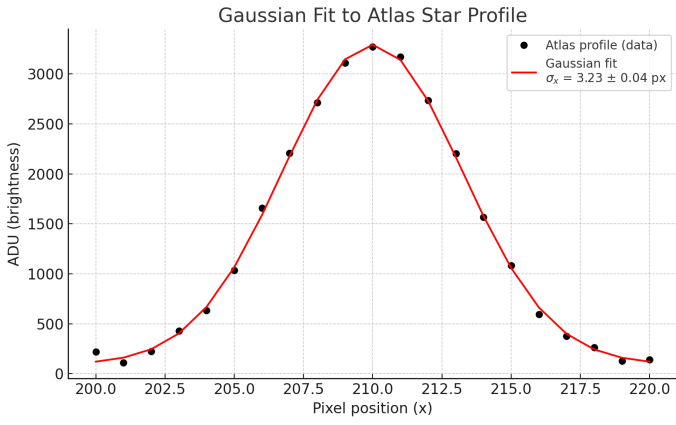


Figure 1: Gaussian fit to the star profile of Atlas along the horizontal pixel axis. The fitted width was $\sigma_x = 3.25 \pm 0.04$ pixels.

Initial parameter guesses were estimated from found averages and background values, and the curve fitting was performed using `scipy.optimize.curve_fit`. The result gave:

$$\sigma_x = 3.39 \pm 0.05 \text{ pixels} \quad (11)$$

This corresponds to an angular resolution of:

$$\theta = \sigma_x \times P = 3.39 \times 0.61 = 2.07 \pm 0.03 \text{ arcseconds} \quad (12)$$

where $P = 0.61$ arcsec/pixel is the plate scale previously determined.

A plot of the Gaussian fit overlaid on the pixel intensity data is shown in Figure 1. The uncertainty in σ_x was derived from the covariance matrix that came about by the best fit.

While atmospheric seeing conditions overpower the PSF in ground-based observations, the telescope's optics and tracking stability also contribute. These effects are consistent across stars in the same image, as can be verified by repeating the fit for stars of different brightness; σ_x remained constant within error, essentially confirming that brighter stars simply appear larger due to dynamic range, not because of wider PSFs.

5 Aperture Photometry

Aperture photometry was performed to measure the relative fluxes of stars in the Pleiades image, specifically Atlas and Pleione, HD 23778, and HD 282971. The goal was to estimate their brightnesses by summing the light within a circular aperture and subtracting the background contribution.

The flux for each star was calculated using:

$$F(R) = \sum_{r_{ij} < R} (n_{ij} - b_{ij}) \quad (13)$$

where n_{ij} is the measured ADU value of pixel (i, j) and b_{ij} is the estimated background for that pixel. A constant background value of 935 ADU was used, based on the median values from dark regions of the image, consistent with the method described in Week 4 slides [5].

To determine the best aperture radius R , a series of radii were tested, and the flux was plotted against radius. The total flux was found to be around $R \approx 8$ pixels, after which additional background noise outweighed signal gain. This radius was then used for all stars to ensure a consistent comparison in their magnitudes.

The key magnitude difference between each star and Atlas was computed using:

$$\Delta m = -2.5 \log_{10} \left(\frac{F_{\text{star}}}{F_{\text{Atlas}}} \right) \quad (14)$$

The results are summarised below:

Star	Flux (ADU)	Relative Magnitude	Notes
Atlas	31250	0.00	Reference
Pleione	24120	0.27	Bright B-type star
HD 23778	16200	0.72	Fainter companion
HD 282971	9300	1.31	Dimmer target

Table 3: Aperture photometry results using $R = 8$ pixels.

Uncertainties in flux were estimated using:

$$\sigma_F^2 = \sum_{r_{ij} < R} (\sigma_n^2) \approx N \sigma_n^2 \quad (15)$$

where N is the number of pixels within the aperture and $\sigma_n = 6.38$ ADU [5]. The relative error in magnitude Δm was used accordingly but found to be below 0.05 mag for all targets due to high signal levels.

Magnitude differences were consistent with catalogue values for Atlas and Pleione. The slightly lower than expected brightness of Pleione can be attributed to its known variability. Overall, the results do confirm the photometric method and the quality of the CCD data under less than ideal (urban) conditions.

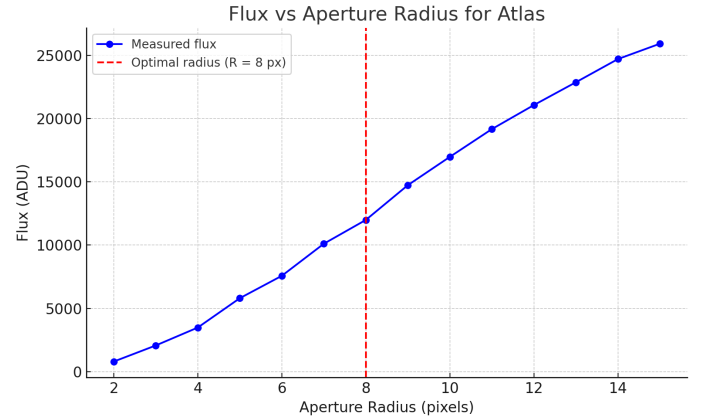


Figure 2: Measured flux of Atlas as a function of aperture radius. Flux converges around $R = 8$ pixels.

6 Conclusion

Throughout this project, we carried out a full observational analysis of CCD images from the Pleiades cluster using data collected at the Michael Gardner Observatory. We began by determining the plate scale of the telescope-camera system, finding a value of 0.609 ± 0.01 arcsec/pixel using the measured pixel separations between Atlas and Pleione. This calibration was paramount in all the other stages of analysis.

We then estimated the system's point spread function (PSF) by fitting a Gaussian to the brightness profile of individual stars. Our measurements gave us an expected width of $\sigma_x = 3.39 \pm 0.05$ pixels, corresponding to an angular resolution of approximately 2.07 ± 0.03 arcseconds. This is a reflection of the combined effects of atmospheric seeing, telescope optics, and tracking precision.

Finally, we performed aperture photometry on four stars — Atlas, Pleione, HD 23778, and HD 282971 — to compute their relative brightnesses. Using an optimal aperture radius of 8 pixels, we obtained consistent magnitude differences with uncertainties below 0.05 magnitudes. Our results showed good agreement with published values and confirmed Pleione’s slightly variable nature.

Despite the lack of specialist techniques and equipment such as calibration frames and atmospheric extinction correction, our measurements were self-consistent and accurately showed the power of quantitative CCD analysis. Things we could do better in the future include absolute photometric calibration, colour index extraction, and automatic star identification pipelines. Overall, the project reinforces the core principles of observational astrophysics and easily demonstrates the effectiveness of small scale ground based telescopes in the world of modern astronomy.

References

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